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ORIGINAL RESEARCH ARTICLE

Hierarchical Architecture of Mesoporous Carbon Wrapped Graphene as Matrix Material of Sulfur for Cathode Material of Lithium-Sulfur Batteries

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Abstract

Hierarchical architecture materials of mesoporous carbon wrapped graphene (MC@G) composite are prepared by a sol-gel coating method, MC@G composites are infiltrated with sulfur to prepare cathode material of lithium-sulfur batteries (Li-S) with high discharge capacity at low current. The MC@G was synthesized using graphene as conductive base material and mesoporous carbon shell wrapped on the surface of graphene as a matrix for loading sulfur, which results in high utilization of active material at low current. The hierarchical architecture of MC@G carbon/carbon nanocomposites forms an effective conducting base material for electron transport in electrode material, which significantly improves the electronic conductivity of cathode material and utilization of active material. Taking advantage of the structure, the S/MC@G cathode material exhibits an initial specific discharge capacity of 1158 mA h g^{-1} and 1131 mA h g^{-1} at current of 0.1 C and 0.5 C, the S/MC@G cathode material exhibits higher capacity retention and rate performance than S/rGO cathode material. We believe that the hierarchical mesoporous architecture of MC@G can also be applicable for designing some other electrode materials for energy storage.

Keywords Lithium-sulfur batteries · graphene · hierarchical architecture materials · sulfur/carbon cathode materials

Introduction

Nowadays, energy storage systems for clean, low cost and high energy density are of great urgency due to consumption of fossil energy, environmental pollution and extensive deployment of portable electronic devices. Therefore, pursuit of advanced energy storage systems with high energy density is critical to meet the demands of key markets.[1–3] Latest, the Li-S batteries have attracted much research attention due to their high theoretical specific energy density of 2600 Wh kg^{-1} and high theoretical specific capacity of 1675 mA h g^{-1}.[4–7] The high specific energy density and high capacity of Li-S batteries make it become a promising choice for next generation of high-energy rechargeable batteries.[8] Furthermore, sulfur is highly attractive as a cathode material for Li-S batteries, because it is environmental friendly, which possesses more advantages over the conventional lithium ion batteries, such as high natural abundance, nontoxicity and inexpensive.[9,10] Despite these advantages, Li-S batteries still face challenges such as serious capacity fading and poor cycle performance, which tremendously hinders their realistic application.[11] Firstly, the poor electronic conductivity of sulfur, intermediate polysulfide products and final discharge products (Li_2S/Li_2S_2) leads to low utilization of active material and limit the high rate capacity.[12] Secondly, lithium polysulfide intermediates (Li_2S_x, $4 \le x \le 8$) formed during the charge/discharge processes, which is soluble in the electrolyte (in organic solvent electrolytes)

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and can shuttle through the electrolyte between anode and cathode during cycling (i.e., that is a "shuttle effect"), resulting in the loss of active cathode material and poor cycle life.[12,13] Thirdly, sulfur cathode would undergo significant volume expansion/shrinkage during lithiation process due to the density difference of sulfur and discharge products, which can lead to the destruction of cathode structure.[14–16] Finally, the shuttle effect of polysulfides is inevitable during charging-discharging, which occurs between lower-order and higher-order forms of polysulfide during the operation of a battery.[17]

The problems mentioned above dramatically affect the development of Li-S batteries, many efforts have been used to prepare sulfur-based composites cathode material, such as novel nanostructured carbon–sulfur composite electrodes,[18] sulfur/conductive polymer composites,[19] sulfur/metal organic framework (S/MOF) composites,[20] sulfur/multi walled carbon nanotubes (S/MWCNTs),[21] sulfur/graphene.[22,23] Besides, nanostructured materials have received tremendous interest due to their unique electrical properties, therefore, many novel design strategies and many novel nanostructured composite materials are conducted to prepare matrix material to encapsulate and disperse sulfur to improve the performance of Li-S batteries,[20,23,24] such as using hollow porous carbon nanosphere (HCNS) as matrix material to encapsulate sulfur prepared HCNS/S composites exhibits excellent electrochemical performance.[25] Using core-shell meso/microporous carbon host for sulfur loading as cathode material can deliver a much higher initial specific capacity and more stable capacity retention than no coating.[26] Designed 3-D hollow foam-like cores to composite with sulfur for cathode of Li-S batteries show high capacity and cycling performance.[27] Combining the advantages of MWCNTs and meso-carbon materials to yield hierarchical architecture MWCNT@Meso carbon composites with sulfur to prepare nanocomposites exhibits high performance matrix for Li-S batteries.[28] Among them, the most widely used method is using sulfur to composite with various carbon materials, by compositing can tremendously improve the conductivity of cathode materials.

Graphene (G) is a one-atom-thick two-dimensional carbon materials, which is considered as a promising candidate for electrode material and substrate material due to its excellent electrical conductivity and remarkable mechanical properties.[22,29] The graphene oxide (GO) is graphene derivatives, which can reduce to graphene by thermal reduction and chemical reduction, and it is well known to that GO can disperse well in water and thus has been routinely described as hydrophilic in the literature, the hydrophilicity of GO has been mainly attributed to the ionizable edge –COOH groups.[30] In terms of chemical process ability of GO and the relationship of GO and graphene, GO has the appeal of being capable of large scales production and is particularly useful for making graphene composites, usually, which is used to a basal material to prepare composite material.[30] Using graphene and reduced graphene oxide as basal materials to composite with sulfur for cathode material of Li-S batteries have been studied by many researchers. However, only using graphene to composite with sulfur is not an effective method to disperse sulfur, although graphene has high conductivity but it may can't confine the polysulfides since it is smooth layered structure, the shuttle effect may easily occur in the cycling. The shuttle effect is a major problem for cycling performance of Li-S batteries, many publications have been put forward that porous structure is an attractive approach to trap polysulfides, many porous carbon materials have been used to composite with sulfur to prepare cathode of Li-S batteries.[31] The Nazar and coworkers demonstrated that a nanostructured sulfur–mesoporous carbon cathode can tremendously improve the cycling performance, which mainly attributes to the advantages of mesoporous carbon with large surface area, high pore volume, interconnected porous structure and high conductivity.[32] Inspired by the characteristic of GO, graphene and porous carbon application in Li-S batteries, we mentioned a new method to prepare hierarchical micro-pores architecture carrier carbon material (MC@G) by combined the advantages of GO, graphene and nanostructure porous carbon materials, which might serve as high performance matrix material for sulfur to prepare cathode materials of Li-S batteries.

In this work, few layered GO was used as substrate material, using a wet chemical method to synthesis SiO_2 particle covers on surface of GO to prepare SiO_2@GO composites, the Poly(furfuryl alcohol) was synthesized and crosslinked with SiO_2@GO composites, and carbonizing the composites to prepare C-SiO_2@GO composites, then by removing SiO_2 particle template to prepare the hierarchical architecture MC@G composites. The MC@G material was used as a matrix material to uniformly encapsulate sulfur to prepare a novel S/MC@G cathode material. The S/MC@G shows high discharge specific capacity at low current and the MC@G material is a promising matrix material for sulfur.

Result and Discussion

The MC@G composite was prepared by using a sol–gel coating method and followed by carbonizing and removing SiO_2 template materials[28,34.] The highly oxidation graphite oxide was prepared by our reported work, the ultrasonic exfoliated method was used to prepare few-layer GO on appropriate ultrasonic condition.[33] The Fig. 1 reveals the different morphology of the materials during the preparation of MC@G composite materials. From the FESEM image of Fig. 1a, the SiO_2 particles is dispersed on the smoothed surface of GO, from the publication we can conclude that

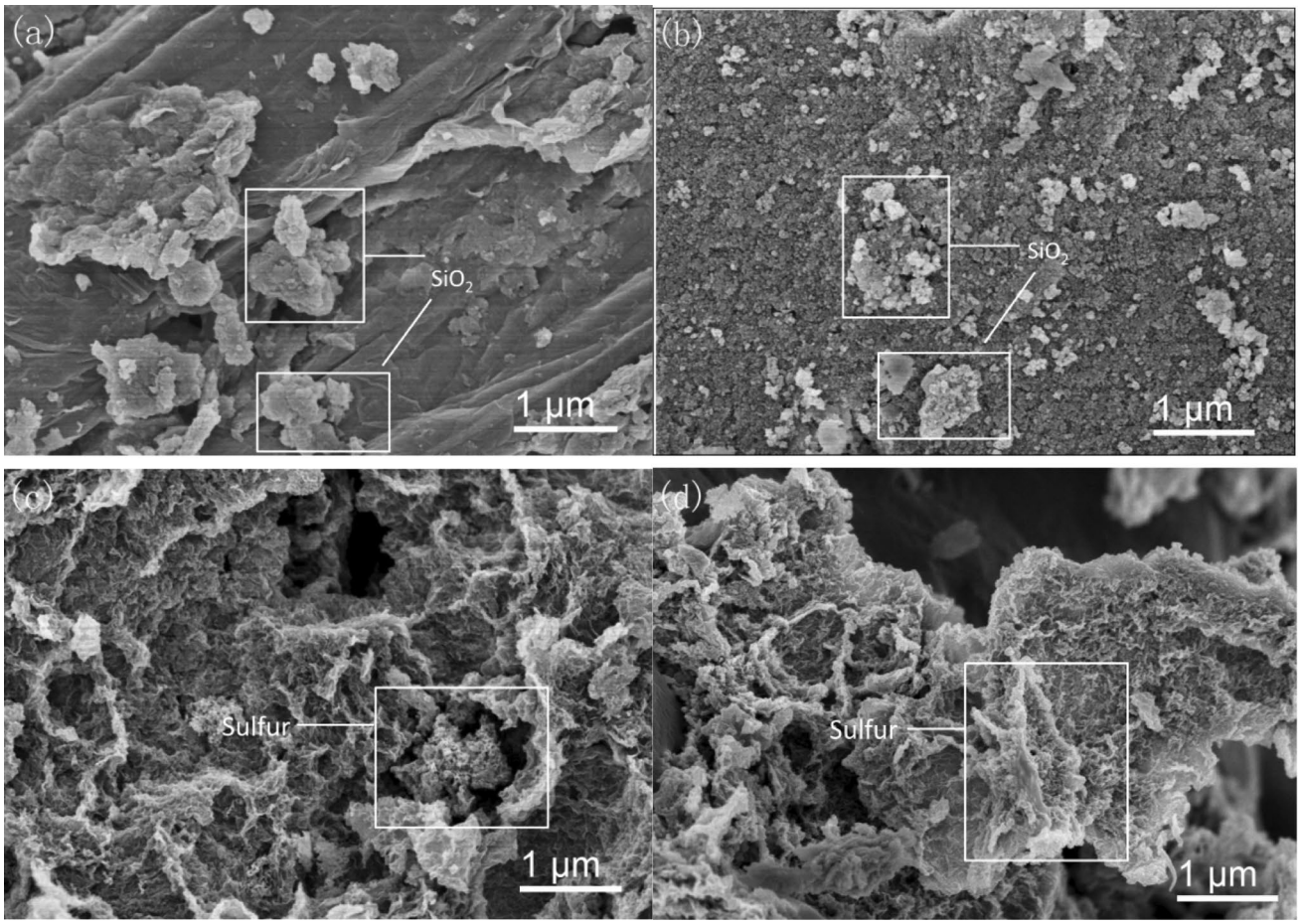

Fig. 1. (a) FESEM images of SiO_2@GO composites; (b) FESEM images of C-SiO_2@G composites; (c) FESEM images of MC@G composites; (d) FESEM images of S/MC@G cathode materials.

the GO sheets are uniformly spreads out in aqueous solution, which because the functional groups of GO not only facilitated the good dispersion of GO in aqueous solution, but also facilitated the interaction between GO and covalent or non-covalent bonds products. The Poly(furfuryl alcohol) was synthesized by acid catalysts, in the synthesis. the oxalic was used as catalysts, after polymerization, Poly(furfuryl alcohol) was crosslinked with SiO_2 particles, the crosslinked composites of Poly(furfuryl alcohol) and SiO_2 particles would interact with functional groups of GO via covalent or non-covalent bonds, which will facilitate the composition of GO and Poly(furfuryl alcohol), the Poly(furfuryl alcohol) crosslinked SiO_2 particles covers on GO obtained P-SiO_2@GO composite materials. After high temperature carbonizing in inert atmosphere, the Poly (furfuryl alcohol) was carbonized to carbon material and most of the functional groups of GO were removed, the GO was reduced to reduce graphene (G), the P-SiO_2@GO converted to C-SiO_2@G. The FESEM images of C-SiO_2@G composites shown in Fig. 1b, the surface of C-SiO_2@G composites are coarse, which is resulted by carbon crosslinked SiO_2 particles and deposited on surface of graphene basal. Compared the FESEM of C-SiO_2@G with GO, the surface of graphene is wrapped with carbon shell, in the MC@G composites, the SiO_2 particles templates were removed by NaOH solution to obtain the MC@G composites, the FESEM of MC@G composites shown in Fig. 1c. Compared the FESEM of SiO_2@GO (Fig. 1a) and C-SiO_2@G (Fig. 1b), it can be concluded that the porous structure is formed in MC@G, the composites are coarse and form a hierarchical architecture. The porous structure MC@G was composited with sulfur to prepare S/MC@G cathode material, the surface morphology shown in Fig. 1d, the FESEM images show that the sulfur coats and immerses into pores of MC@G composites, the coarser surface structure is reserved in S/MC@G cathode materials, which might confines the intermediate lithium polysulfides and attenuates its dissolution during the cycling process, which might facilitates the improvements in the cycling stability of S/MC@G cathode materials.

The element type and element content of SiO_2@GO and S/MC@G composites were measured by energy dispersive spectroscopy (EDS) surface scanning, the results show in

Fig. 2. In SiO_2@GO composites, the mainly element composition are consisted of carbon (C), silicon (Si) and oxygen (O), shown in Fig. 2a. The different scanning areas with different element composition, the atomic composition proportion of Si and O are 27.74 and 72.26, which can be attributed to SiO_2 particles, and the atomic composition proportion of C, Si and O are 6.20, 25.13 and 68.67, which can be attributed to the composites of SiO_2 and GO, from the results of EDS, it can be seen that the SiO_2 particle is distributed on GO. The mainly element composition of S/MC@G are sulfur (S) and C, shown in Fig. 2b, the atomic composition proportion of C, S and O are 79.53, 14.71 and 5.75, the percentage of element weight are 62.88, 31.06 and 6.06, which indicates that sulfur is uniformly distributed with MC@G.

The morphological variation and structure evolution of GO, MC@G and S/MC@G were further corroborated by the FETEM images in the preparation, shown in Fig. 3. The GO is layered structure and the size are 500–1000 nm, which is parallel spread out in aqueous solution and almost transparent state, after compositing with mesoporous carbon, it is obvious that the microstructure of the mesoporous carbon covering on the graphene sheets, the surface of graphene forms the coarse and porous structure, contrasting the Fig. 3a, b, c and d can clearly prove the difference of GO and MC@G, a large number mesoporous spread out on layered graphene in the MC@G, the coarse and porous structure of MC@G will be beneficial to loading of the sulfur, which will dramatically improve the amount of sulfur trapped onto the structure. In the S/MC@G, the sulfur deposition and penetrate into the inner mesoporous of MC@G through the outer mesoporous shell, as shown in Fig. 3e and f it is visible evidence for the sulfur was confined within the carbon shell.

Sulfur was directly melt-absorbed onto the surface of carbon shell, the FETEM results is consistent with the FESEM observations. By encapsulating and penetrating sulfur into the inner mesoporous structure of MC@G will keep the sulfur with highly dispersed state, which ensures the sulfur can fully contact with conductive carrier material and electrolyte. Through the combination of graphene and mesoporous carbon will facilitate the composition of sulfur and conductive carrier material, which will not only serve as a host to trap sulfur/polysulfides, but also provides a pathway for Li^+ transports. Therefore, designing hierarchical architecture of MC@G might is an ideal carrier material to prepare sulfur cathode enhancing the performance of Li-S batteries, which can improve the structural stability of cathode by accommodating the volume expansion of sulfur during discharge and charge process.

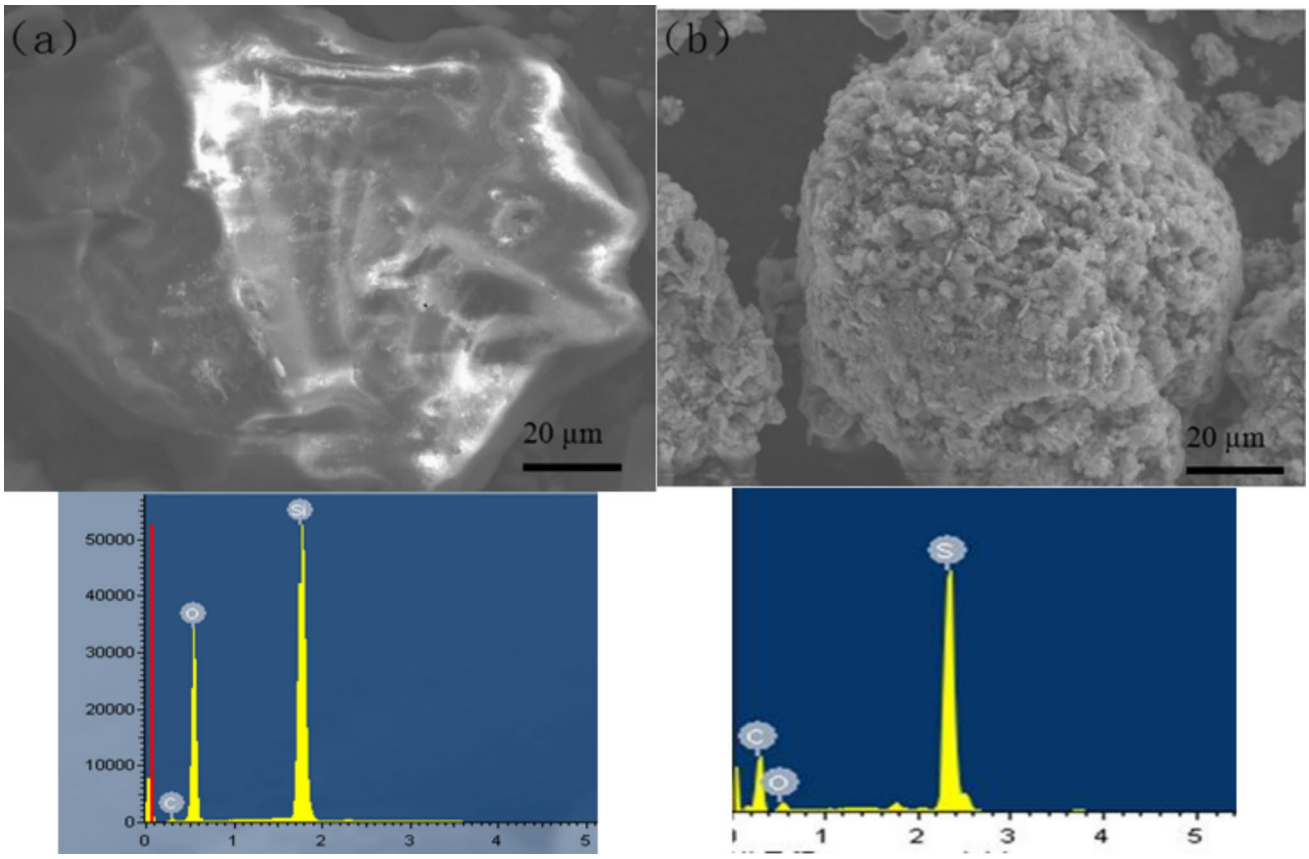

Fig. 2. (a) EDS images of SiO_2@GO composites; (b) EDS images of S/MC@G cathode materials

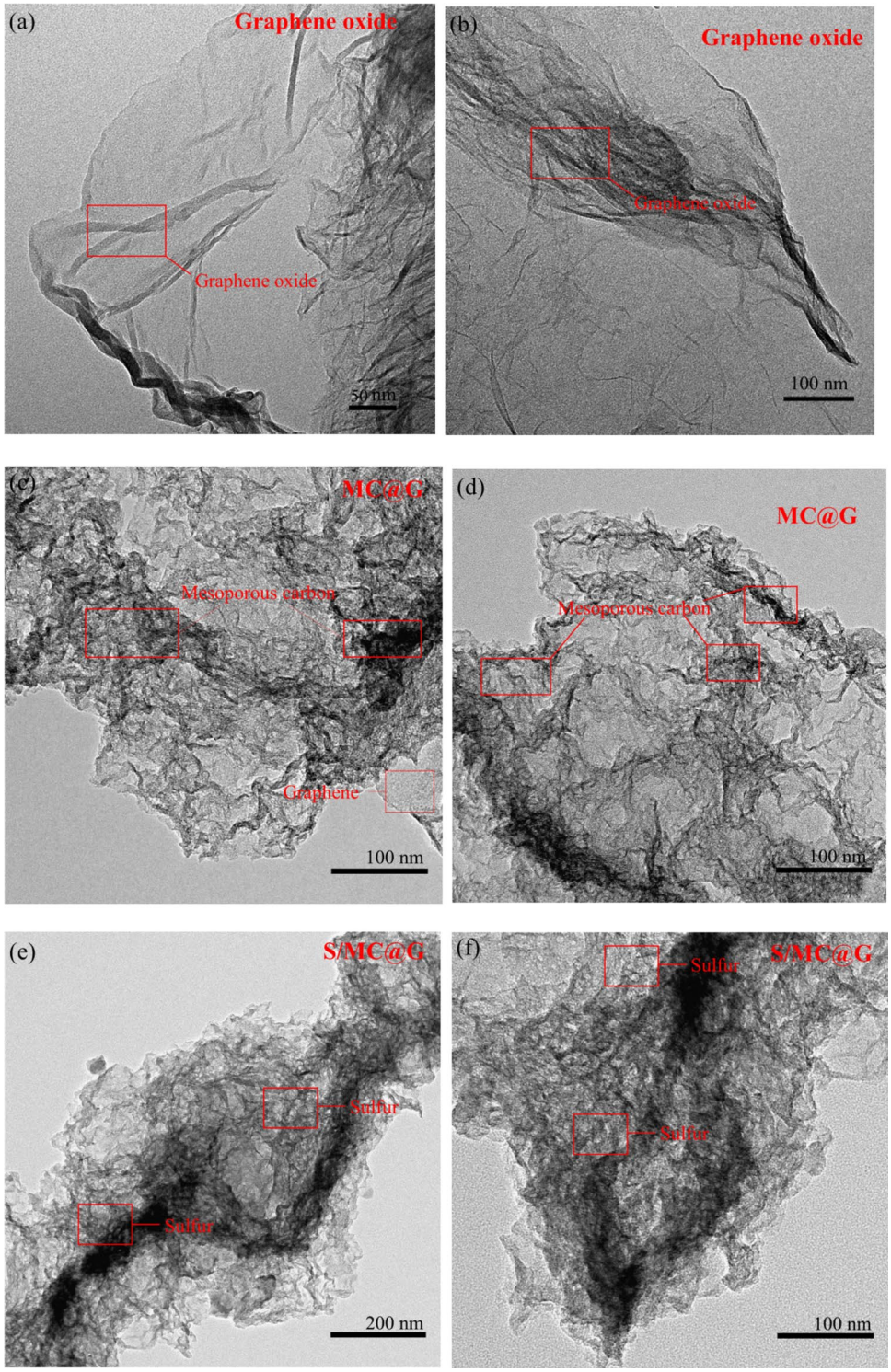

Fig. 3. (a, b) FETEM images of exfoliated graphene oxide on ultrasonic power of 150 W for 180 min; (c, d) FETEM images of MC@G composites; (e, f) FETEM images of S/MC@G cathode materials.

N_2 adsorption-desorption isotherms and the pore-size distribution of rGO, MC@G and S/MC@G cathode materials shown in Fig. 4a and b, which will supply directly evidence for sulfur encapsulates into porous of MC@G. In Fig. 4a, rGO and MC@G have a specific surface area of 449.3 $m^2\ g^{-1}$ and 272 $m^2\ g^{-1}$, adsorption average pore diameter are 18.14 nm and 4.2 nm, the difference between rGO and MC@G is that the rGO was covered by mesoporous carbon formed MC@G. In the N_2 adsorption-desorption isotherms of rGO and MC@G, there are hysteresis loops at a relative pressure of P/P_0 between 0.4 to 0.8, which is characteristic of mesoporous structure according to the

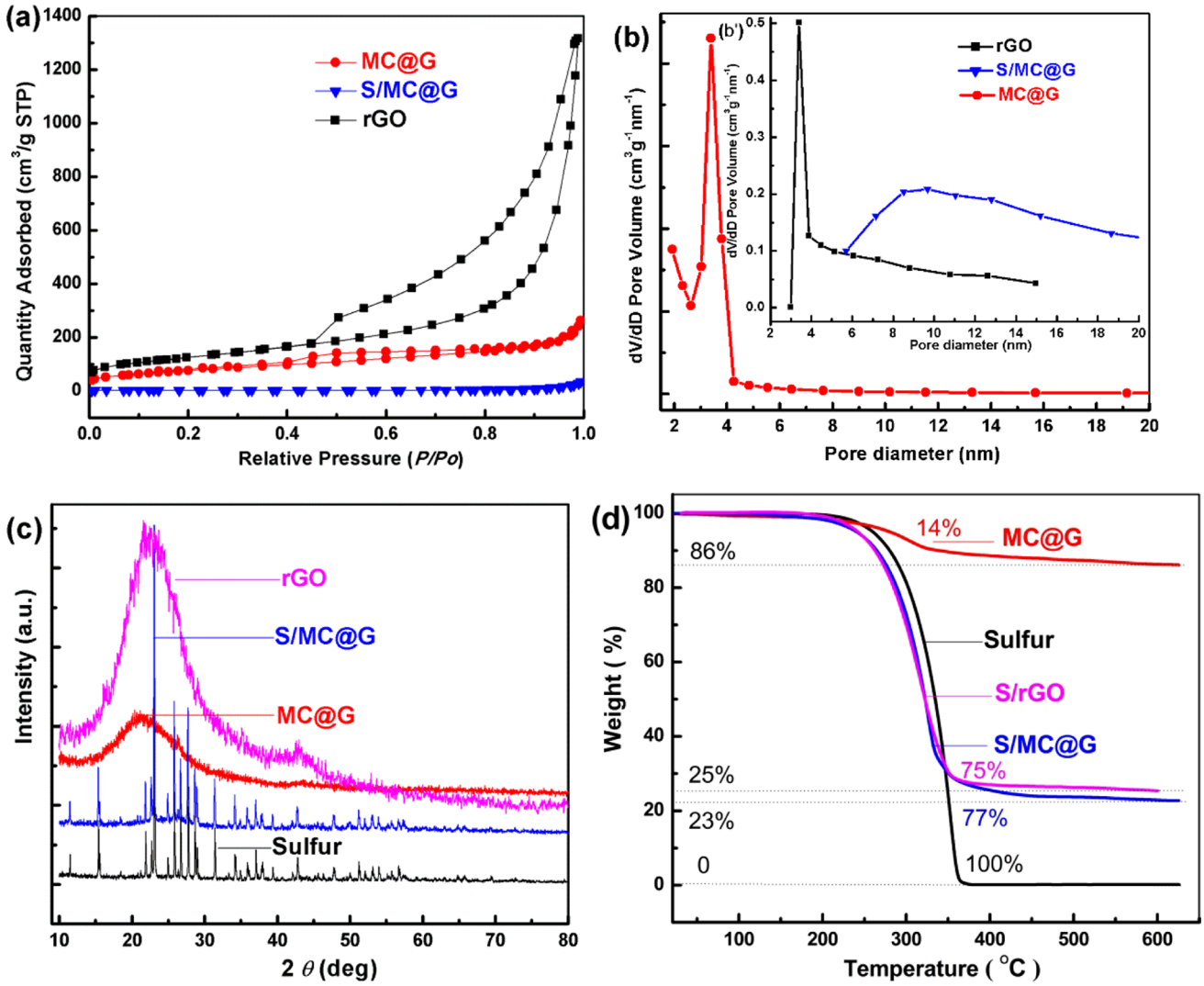

Fig. 4. (a) Nitrogen adsorption–desorption isotherms of rGO, MC@G and S/MC@G; (b) pore-size distribution plots of rGO, MC@G and S/MC@G; (c) XRD patterns of sulfur powder(sulfur), rGO, MC@G and S/MC@G; (d) TGA curves of sulfur, MC@G, S/rGO and S/MC@G.

publication.[28] After sulfur infiltrated into MC@G, the BET surface area dramatic decrease, the BET surface area of S/MC@G is only 8.4 m^2 g^{-1}. In Fig. 4b, the BJH desorption pore volume distribution of rGO and MC@G is 3–4 nm, the coated of mesoporous carbon and sulfur infiltration caused the small porous of rGO and MC@G, the mesopores can be observed from the pore size distribution, and the porous distribution of MC@G are inhomogeneity in the FETEM image, which is in connected with distribution of the SiO_2 particles template, removal of templates renders the cavity is reserved formed the mesoporous structure. From the pore-size distribution of S/MC@G, the big mesoporous is reserved, according the pore-size distribution of S/MC@G, the S/MC@G shows little porous, the BET average pore diameter of S/MC@G is 9.1 nm, the porous size of the S/MC@G interior is mostly around 10–30 nm, which indicates that sulfur have penetrated into the micropores of MC@G composites, and the big mesoporous structure is still reserved in local region of S/MC@G. The porous structure of S/MC@G may act as polysulfide reservoirs and hold sulfur, which are favorable for high sulfur utilization and can confine or alleviate the part of shuttle effect of Li-S battery in the cycling.[28] The results of N_2 adsorption isotherms and the pore-size distribution are consistent with the observed of FESEM and FETEM.

The x-ray diffraction (XRD) patterns of MC@G, S/MC@G, sulfur powder and rGO are shown in Fig. 4c. The GO is reduced by thermal reduction in carbonization, in XRD pattern of rGO, a diffraction peak appear at around $2\theta = 22.5°$, which can be attributed to the (002) reflections of thermal reduced graphene.[33,35] It means that the rGO with amorphous characteristic and most of the oxygen containing

functional groups of GO are efficiently removed by fast thermal exfoliation. The XRD pattern of MC@G is similar to the XRD pattern of rGO, which indicates the MC@G with amorphous carbonaceous materials characteristic, the composition of MC@G are amorphous characteristic carbon. In order to prove the sulfur has composited with MC@G, the XRD of sulfur powder, MC@G and S/MC@G composites are compared, the S/MC@G shows similar sharp and strong sulfur diffraction peaks, which can prove that the MC@G and sulfur is composited well, the sharp diffraction peaks denote that sulfur well-defined crystal structure, while the broad diffraction peak appears at about 22.5°, which indicates the amorphous characteristic of MC@G is reserved. In the XRD pattern of S/MC@G exhibits the diffraction peaks characteristic of MC@G and sulfur, which further demonstrates that sulfur exists as a highly dispersed state and it is composited well with MC@G, the XRD result is consistent with the conclusion from FESEM, FETEM analysis.

In order to determine the sulfur content in S/MC@G cathode materials, thermogravimetric (TGA) analysis was performed under N_2 atmosphere, the TGA curve depicts the weight losses of sulfur, MC@G, S/rGO and S/MC@G, through the comparison of TGA curve, we can conclude that most of losses are correlated to the sublimation of sulfur. According the Fig. 4d, the weight loss of the sulfur powder starts at about 200 °C and stops at about 370 °C, which is ascribed to the sublimation of sulfur. The composite of S/MC@G and S/rGO exhibits the similar TGA curve with sulfur powder, through the comparison between the TGA curves of sulfur powder, MC@G, S/rGO and S/MC@G, we can work out the sulfur content of S/MC@G and S/rGO cathode materials, the actual sulfur content of S/MC@G can express as 77–14% and the actual sulfur content of S/rGO can expressed as 75–14%. Thus, the actual sulfur content of S/MC@G and S/rGO are 63 wt.% and 61 wt.%.

To investigate the electrochemical performance of S/MC@G-63% cathode materials, the CR2025 coin cells were assemble in glovebox by using S/MC@G-63% and S/rGO-61% as cathode material, metallic lithium sheets as counter electrode, acetylene black as additive and a PVDF as binder. Galvanostatic charge–discharge measurements are evaluated in a voltage window between 1.5 V and 3.0 V at room temperature. All of the capacity values of the cathode materials are calculated based on sulfur mass. The charge/discharge curves of S/MC@G cathode material at different current density are shown in Fig. 5a, b and c, the charge/discharge curves of S/rGO cathode material at current density 0.1 C are shown in Fig. 5d. The initial discharge capacity of S/MC@G cathode material is 1158 mA h g^{-1}, at the current rate of 0.1 C, after the initial cycle, the composite shows a good electrochemical reversibility with the reversible capacity of about 1107 mAh g^{-1} and 1027 mAh g^{-1} at the 2nd and 5th cycle. At current density of 0.5 C, the initial discharge capacity are 1131 mA h g^{-1}, after the initial cycle, the reversible capacity of about 1067 mA h g^{-1} and 983 mAh g^{-1} in the 2nd and 5th cycle. The S/MC@G cathode material at higher current of 2 C shows the initial discharge capacity of 600 mAh g^{-1} and reversible capacity of 574 mA h g^{-1} are presented after 5cycles, the S/MC@G cathode material exhibits sloping discharge/charge plateaus and the plateau even disappears when the current density raised to 2 C rate, and the voltage polarization is much higher, indicating the slow redox reaction kinetics, according to the publication.[36] As a comparison, the S/rGO cathode material, which shows high initial discharge capacity than S/MC@G cathode material at 0.1 C current density, the initial, 2nd and 5th discharge capacity are 1257 mA h g^{-1}, 1171 mA h g^{-1} and 932 mA h g^{-1}, though the initial discharge capacity is obviously higher than S/MC@G cathode material, however, from the comparing of capacity decreasing tendency, we can speculate the S/rGO cathode material with low capacity retention, which can be verified in Fig. 6a. From the other reported values with high energy Li-S batteries, the cycling performance of the S/MC@G and S/rGO cathode material is related to the Electrochemical impedance spectroscopy, respectively relating to charge-transfer resistance (R_{ct}) and the mass-transfer process, the hierarchical porous structure can mitigate the dissolution of lithium polysulphides and deposition of Li_2S/Li_2S_2, and after cycling, the R_{ct} of S/rGO cathode material will increases significantly, which exhibited poor capacity retention.[36–39]

It can be observation from Fig. 5a, b, c and d, there are two discharge potential plateaus and one charge potential plateaus in all discharge/charge profiles of S/MC@G cathode material, which is typical discharge and charge curves of the sulfur–carbon cathodes. There are two typical discharge plateaus at 2.05 V and 2.35 V in the discharge curves of S/MC@G cathode materials at current of 0.1 C and 0.5 C. At high current of 2 C, the discharge potential plateaus decrease to 2.18 V and 1.7 V, the two potential plateaus are according to the typical reaction of sulfur with lithium during the discharge process. Although the overall sulfur redox reaction is simply given as $S_8 + 16Li^+ + 16e^- \rightarrow Li_2S$, but the mechanism is known to be very complicated. The reaction involves a multistep electrochemical process associated with disproportionation of reduction intermediates of sulfur, and lithium polysulfide species (Li_2S_m) with different chain lengths. The redox mechanism is even more complicated, the stability (or solubility) and reactivity of Li_2S_m strongly depend on the electrolyte properties such as polarity, composition, and salt concentration.[28,40] According the literature, the first plateau from 2.35 V to 2.05 V is contribute to a minor part to the overall capacity, which is corresponding to formation of long chain polysulfides (Li_2S_x, $4 < x < 8$),[28,41] The second plateau at 2.05 V is the conversion of polysulphides to Li_2S_2 and then to Li_2S, which is accompanied with the consumption

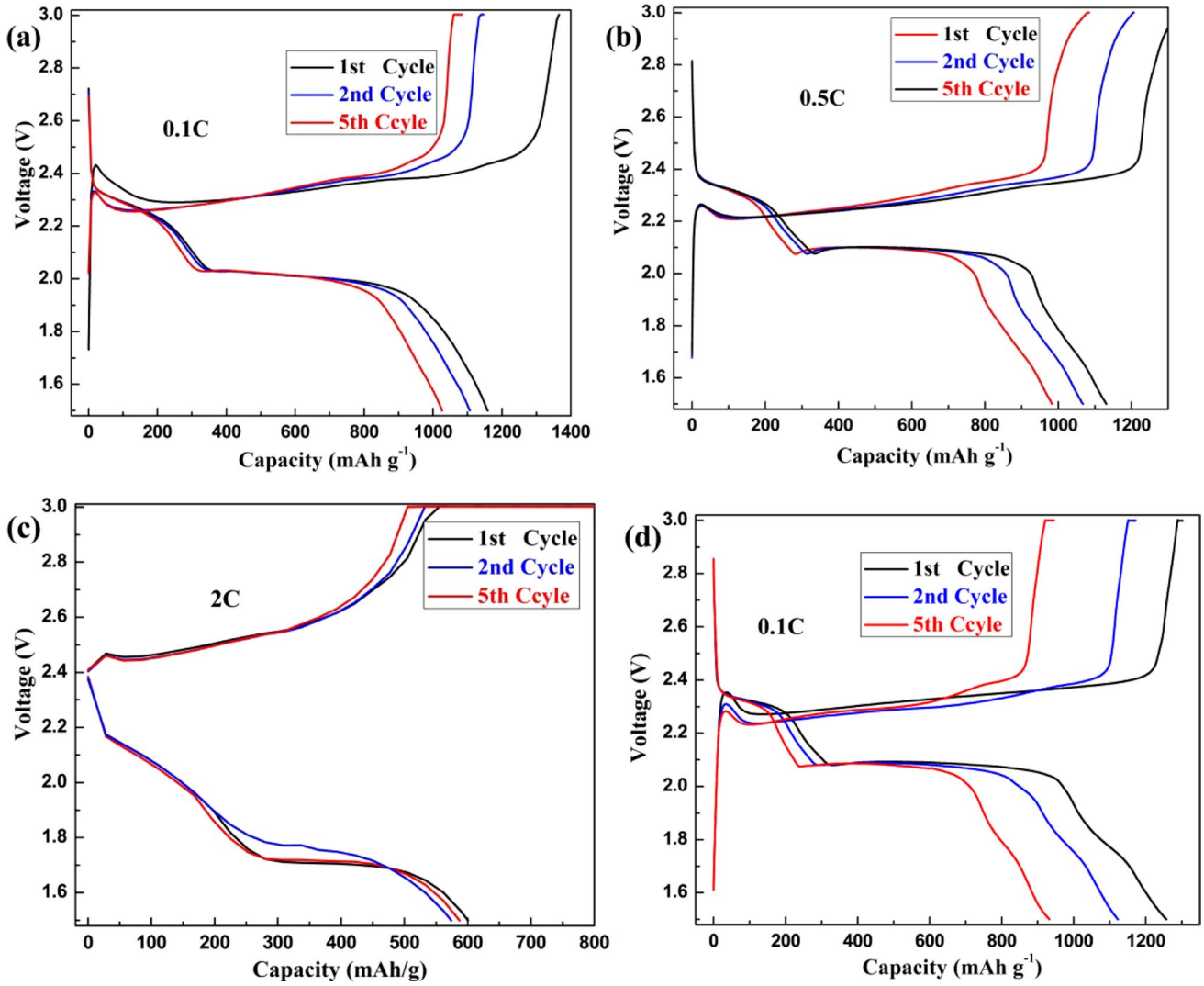

Fig. 5. (a) Charge/discharge voltage profiles of S/MC@G-63% at current of 0.1 C (b) Charge/discharge voltage profiles of S/MC@G-63% at current of 0.5 C; (c) Charge/discharge voltage profiles of S/MC@G-63% at current of 2 C; (d) Charge/discharge voltage profiles of S/rGO-61% at current of 0.1 C.

of longer polysulfides and a concomitant increase in shorter polysulfides, the higher-order polysulphides is reduced to lower-order polysulphides (Li_2S_x, $2 \leq x \leq 4$) at the lower voltage plateau, the sudden potential drop at the end of the second plateau can be observed, which is corresponding to the conversion of all reducible polysulfides to Li_2S precipitates.[40,41] Commonly, the reaction kinetics of 2.35 V plateau is faster than 2.05 V plateau, thus the discharge plateau of 2.35 V is shorter than discharge plateau of 2.05 V,[42,43] the charge process, the long oxidation plateau represents the conversion from lithium sulfides to polysulphides and sulfur.[43–45] In the cycles process, the discharge capacity decay seems to be inevitable according the literature, it is resulted by serious shuttle mechanism of the soluble polysulfide. It is obvious that in the initially cycles, the charge capacity is higher than that of discharge capacity, which dues to the high mobility of the soluble polysulfide, creating a polysulfide shuttle phenomenon during charge process. With the increase of cycle number, the solubility of polysulfide will decrease, since the density gradient of soluble polysulfide increase in electrolyte, the insoluble and soluble products tend to balance. Clearly, two discharge potential plateaus with a hysteretic slope could be observed at the 2 C, the electrochemical mechanism of the sulfur cathode result soluble lithium polysulfides were produced in the upper potential plateau and insoluble reduction products (Li_2S_2 and Li_2S) were alternately formed in the lower potential plateau, the reaction kinetics of high potential plateaus is faster than low potential plateaus, therefore, at high rate, long chain polysulfides are produced and the capacity of the cathode

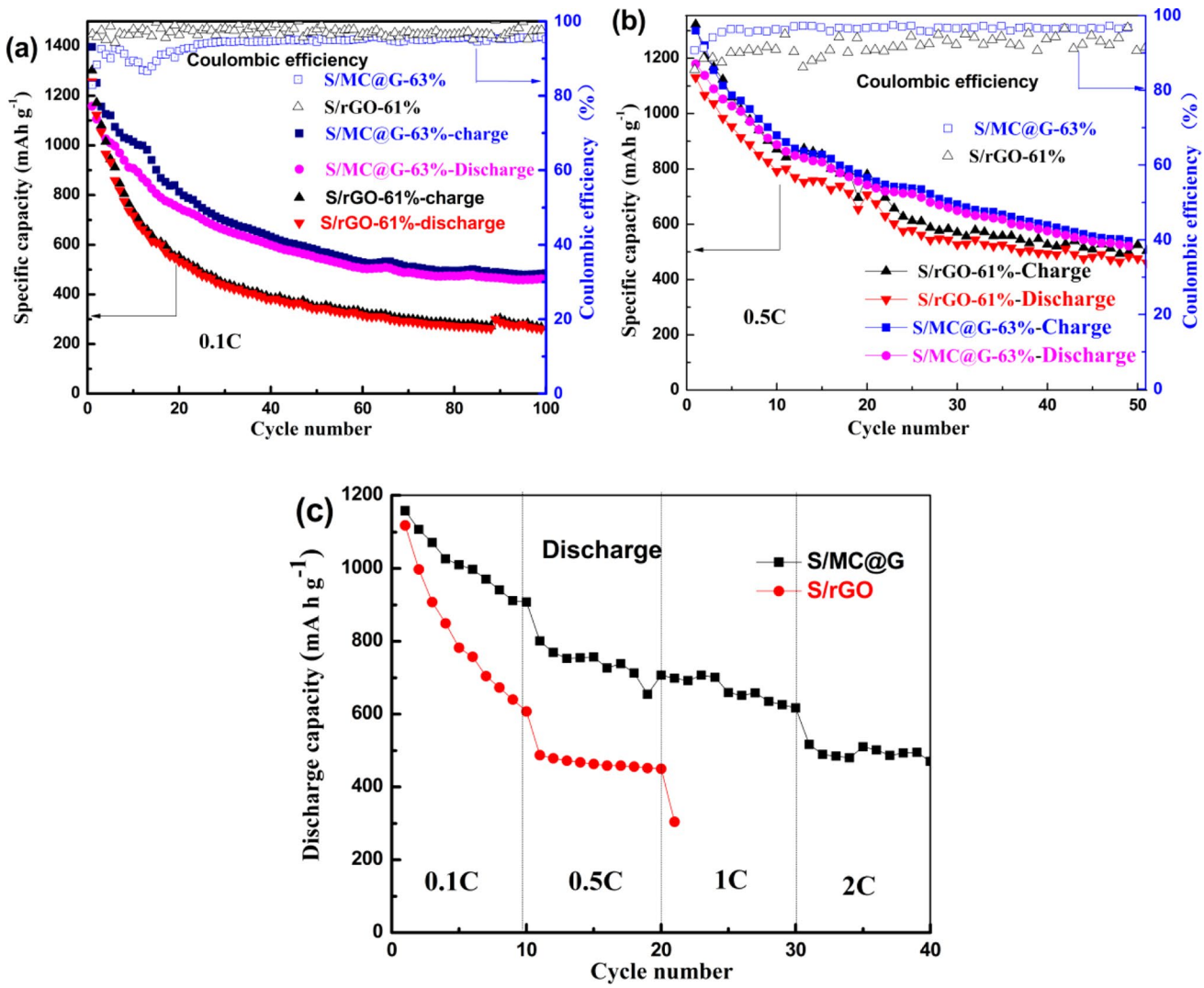

Fig. 6. (a, b) Cycle performance of S/MC@G-63% and S/rGO-61% cathode materials at a current density of 0.1 C and 0.5 C; (c) The variation in discharge capacity with cycles of S/MC@G-63% and S/rGO-61% cathode materials at different current densities.

material is major contributed by the high potential plateaus, and little parts of capacity of cathode is contributed by the low potential plateaus, the reaction of higher-order polysulphides are reduced to lower-order polysulphides at low potential plateaus is not complete, at same time in the charge process, the Li_2S_2 and Li_2S is oxide to higher-order polysulphides is not complete at high rate which results polarization obtained without a delay.[44–47]

The cycling stability and rate performance are another key aspect to limiting the practical application of lithium-sulfur batteries. The cycling performance and rate performance of S/MC@G and S/rGO-61% cathode materials are shown in Fig. 6a and b, which is the cycling performance of S/MC@G-63% and S/rGO-61% cathode materials respectively. At current of 0.1 C, the initial discharge capacity of S/MC@G-63% is 1158 mA h g^{-1}, after 100 cycles, which has 45% retention of initial capacity, however, S/rGO-61% cathode material, the initial discharge capacity is 1257 mA h g^{-1}, after 100 cycles the capacity retention only 20% of initial capacity, more excellent performance of S/MC@G cathode material implies that the majority of the sulfur/polysulfide intermediates were immobilized by MC@G. At current of 0.5 C, the initial discharge capacity of S/MC@G-63% cathode material is 1131 mA h g^{-1} after 50 cycles, the capacity retention is around 50%. However, S/rGO-61% cathode material have the initial discharge capacity is 1179 mA h g^{-1}, after 50 cycles the capacity retention is only around 44%. From the compare of capacity retention between S/MC@G-63% and S/rGO-61% we conclude that the MC@G was used to matrix material of sulfur exhibits more excellent discharge capacity and capacity retention at low rate than rGO, which is attributed the porous structure,

and the meso-carbon coating on graphene play an important role, using reduced graphene as basal material can improve the conductivity of cathode materials, the porous structure of MC@G can effectively disperse sulfur into the porous structure, which can improve the conversion efficiency of sulfur, thus, in the low current rate, the S/MC@G cathode materials exhibits excellent performance. In our work, hierarchical architecture MC@G composites with the high conductivity and excellent porosity have been proved, due to the strong confinement of meso-carbon and graphene, sulfur in forms of small nanoparticles disperses into porous of MC@G. The hierarchical architecture of MC@G is expected to be promising matrix material to disperse the sulfur, S/MC@G cathode material form multiple active sites can greatly suppress active material loss, the rGO as substrate layer, which provide an effective conducting base material for ion transport, MC covered on rGO, the sulfur was infiltrated into the porous, which can improve electrolyte permeability, enabling fast reactions between Li^+ ions and sulfur, besides, the MC/G with hierarchical architecture can accommodate the volume expansion and entrap sulfur/lithium polysulfides.[37] Thus, the cathode materials with hierarchical architecture are beneficial to high performance lithium-sulfur batteries science the hierarchical architecture can accommodate the structure stability of cathode during the cycling.

In the initial charge/discharge cycling of S/MC@G cathode materials, the coulombic efficiency of the coin cell is around 84%, after 10 cycles, the coulombic efficiency increased to above 90%. After 20 cycles, the whole coulombic efficiency reached a stable level around 95%, in the initial cycling, the low coulombic efficiency is due to the polarization of the battery is serious in the initial cycling. According the literature, the viscosity of the electrolyte reaches the maximum value because of the dissolution of lithium polysulfides in the electrolyte, which can increase the transfer resisitance of ion and increases the polarization of the battery.[36,38,39,45] In the initial cycles, the redistribution of active substance is inevitable because of the dissolution of lithium polysulfides during the charge/discharge process, according the publication, the high-rate capability and the cycle stability are mainly related to the interfacial charge-transfer process and lithium-ion diffusion in the composite due to the inescapable aggregation of insulated Li_2S on the cathode surface during cycling.[36,40,43,44] The S/rGO cathode material shows relatively poor cycling stability than S/MC@G cathode material, we speculate that the part of sulfur deposition on the surface of rGO, the discharge products expose to the electrolyte, and resulting serious shuttle effect, implying low sulfur utilization with severe polysulfide dissolution, the irreversible products were reserved on the surface of rGO along with polarization of Li-S battery, thus, in the Fig. 6b it is very obvious that the coulombic efficiency of S/rGO is only about 90%, the coulombic efficiency of S/MC@G is approximately 100% during cycling. The S/rGO cathode material exhibits more severer polarization and sluggisher redox reaction in contrast with S/MC@G cathode material since the different between the structure especially in high rate, and the MC@G with hierarchical pores structure can part confine or alleviate the shuttle effect of Li-S battery in the cycling.[36,40]

Figure 6c exhibits the rate performance of S/MC@G and S/rGO-61% cathode material. With the increase of current density from 0.1 C to 2 C, the specific discharge capacity of S/MC@G cathode material gradually decrease, the current density at 0.1 C, 0.2 C, 0.5 C and 2 C, the average discharge capacities during 10 cycles are 1010 mA h g^{-1}, 737 mA h g^{-1}, 664 mA h g^{-1}, 493 mA h g^{-1} respectively, the discharge capacities is stabilized. However, the S/rGO-61% cathode material displaying a very poor rate capability, and at high current density of 1 C and 2 C, the battery will unable to continue to discharge. Thus, using MC@G as a matrix material of sulfur to prepare cathode material exhibits excellent rate performance than only using rGO composite with sulfur to prepare cathode material of Li-S battery. The more excellent rate capability of S/MC@G cathode material is attributed to the hierarchical porous structure, which provides more efficient electrical conductivity and electronic/ionic transport interfaces than S/rGO cathode material. From the comparison of electrochemical performance between S/MC@G and S/rGO cathode material, we can conclude that porous structure is a useful strategy to improve the stability of cathode, which is beneficial for trapping the polysulfide and accommodate volume expansion of sulfur and polysulfide during discharge and charge process. On the other hand, hierarchical porous structure can improve the conversion efficiency of active material by improving the dispersion and conductivity of S/MC@G cathode materials.

Conclusion

In summary, the hierarchical architecture mesoporous carbon wraps graphene (MC@)G) composite materials are prepared by sol–gel coating method and template method. Taking the advantage of graphene oxide, graphene and mesoporous carbon, designed a hierarchical structure carbon material, the MC@G was used as matrix material to infiltrating sulfur prepared S/MC@G cathode materials. The S/MC@G cathode material exhibits high initial discharge capacity of 1158 mA h g^{-1} at a current density of 0.1 C, 100 cycles with the capacity retention are 45%. At current of 0.5 C, the initial discharge capacity is 1131 mA h g^{-1}, 50 cycles with the capacity retention are 50%. Using mesoporous carbon wraps graphene for matrix material of sulfur can improve the performance of cathode materials of Li-S batteries, which shows more excellent

performance than S/rGO cathode materials. Graphene as the conductivity material provides sustaining intimate electrical contact and provides electronic pathways, using mesoporous carbon covers on graphene as the matrix to disperse the sulfur is beneficial to the utilization of active material and hierarchical pore structure can decrease the shuttle affect to capacity fading, which exhibits excellent rate performance than rGO. Our work offers a novel design of mesoporous carbon materials with hierarchical architecture, the same strategy adapted in this study could be helpful to explore and develop other electrode materials with a poor electric conductivity and dispersion, the insights and novel concepts of building mesoporous carbon wraps graphene composite materials will be valuable for future improving the performance of cathode material in lithium-sulfur batteries.

Material and Methods

Synthesis of Mesoporous Carbon-Wrapped Graphene Composite Material

The MC@G was achieved by using few layered GO as substrate material, in the GO solution, SiO_2 particle was synthesized and covered on the surface of GO to form the SiO_2@GO composite structure. In a typical synthesis, 1.8 g of GO was dispersed in 400 ml ethyl alcohol (EtOH)–H_2O (1:1 v/v) mixture solution, the GO was prepared according our publicized work, specifically, at 40 °C, the graphite oxide was dispersed and exfoliated with ultrasonic at ultrasonic power 150 W for 180 min to prepare fewer layered GO.[33] After sonicating, 4 ml of ammonia hydroxide (28 wt.%) was added followed, then dropwise addition of tetraethyl orthosilicate (TEOS) (36 mL in 80 mL EtOH) with peristaltic pump and the mixture solution was stirred at 40 °C for 24 h obtained the SiO_2@ GO composites.

The Poly(furfuryl alcohol) was synthesized as the carbon source in SiO_2@GO solution. Typically, 15 g Furfuryl alcohol and 0.4 g oxalic acid was added into SiO_2@GO solution. After sonicating for 40 min, the mixture solution was maintained at 80 °C for 48 h, afterwards, the mixture solution was dried at 80 °C for 12 h. The Poly(furfuryl alcohol) cross-links SiO_2 particle coats on the surface of GO (P-SiO_2@GO), P-SiO_2@GO composite was pyrolyzed at 150 °C for 3 h and carbonized at 900 °C for 2 h to prepare carbon-coated SiO_2@graphene (C-SiO_2@G) composites. Finally, the silica of C-SiO_2@G was etched in a NaOH solution to obtain MC@G composites. The GO was reduced by thermal reduced at 900 °C for 2 h to prepare reduced graphene (rGO).

Infiltrating Sulfur into MC@G Composites to Prepare Cathode Material of Li-S Batteries

Infiltrated sulfur into MC@G and rGO to prepare cathode material of Li-S batteries by using simple melt-diffusion strategy. Typically, sulfur powder and MC@G composites were mixed at the weight ratio of 3:1, sulfur powder and rGO were mixed at the weight ratio of 3:1, mixtures were placed into a sealed vessel and set the temperature at 155 °C for 24 h under an argon atmosphere to obtain S/MC@G and S/rGO cathode material. Consequently, the slurry was prepared by ball milling 80% cathode material, 10 wt.% ketjen black (KB), 10 wt.% poly(vinylidene fluoride) (PVDF) mixed in *N*-methylpyrrolidone (NMP) solvent dispersant. Then, the resulting slurry was coated on an aluminum foil using a doctor blading method and the thickness of slurry 0.1 mm, the diameter of the pole piece is 12 mm. Electrodes were dried at 45 °C under vacuum for 12 h. CR2025 type coin cells were assembled in a glove box with oxygen and water contents less than 0.1 ppm, lithium metal sheets were used as counter electrode and porous membrane (Celgard 2400) as the separator, the electrolyte was adopted mixture of 1 M lithium bis(trifluoromethanesulfonyl) imide (LiTFSI), 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) (v: v = 1: 1), with 0.1 M lithium nitrate ($LiNO_3$) inside.

Characterization

The prepared SiO_2@GO, C-SiO_2@G, MC@G and S/MC@G were characterized by field emission scanning electron microscopy (FESEM, Tecnai G2 F20). GO, MC@G and S/MC@G were characterized by Field Emission Transmission Electron Microscopy (Tecnai G2 F20, FETEM). x-ray diffraction (XRD) patterns were obtained using D8 ADVANCE diffractometer equipped with Cu Ka radiation ($\lambda = 1.5406$ Å) to characterize the thermal reduced graphene (rGO), sulfur powder, MC@G and S/MC@G. Thermogravimetric analysis (METTLER TGA/DSC 1 SF/1382) was conducted to determining the sulfur content of S/MC@G. Nitrogen adsorption-desorption measurements were performed by the Brunauer–Emmett–Teller (BET) method and porosimetry analyzer (Micromeritics ASAP2020). The charge–discharge curves were tested by using NEWARE battery test system in a voltage range of 1.5–3.0 V, electrochemical measurements were performed at room temperature. The specific capacities at different current density were calculated based on the mass of sulfur alone.

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Data Availability Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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